Design Brief – Review 2

Form BM01. Sections 0–8: maximum 4 A4 pages (font 11, margins 2 cm). Appendix A is not counted. Remove all grey guidance text and example rows, then export to PDF. File name: SWP490_<ProjectCode>_R2_DesignBrief.pdf

0. Header

Project code	FA26SE183
Project name	VISTA: Vision-Language-Action System for Tabletop Manipulation with Dual Robot Arms
Team	GFA26SE94
Supervisor	Lâm Hữu Khánh Phương
Repository link (tag review-2)	https://github.com/SKYNET-Thesis/se (tag review-2: pending)
Version	Registration v1.2 (proposed)
Date	27/10/2026

1. Scope

1.1 Registration Changes

Summary only; details in Form BM03 (Record of Changes). Write "No changes" if none.

Section	Current	Proposed	Reason
Scope	Earlier approved version: not available to compare	Registration v1.2 — four school-to-experiment learning workflows	Review 1 scope alignment; requires verified old version
Robot execution	Hardware E2E proof not supplied	Schematic active in Education API; physical task still target	Must not present planned as implemented

1.2 Out of Scope

Item	Reason
Standalone mobile	Outside mandatory Education Platform Review 2 commitments
Training VLA foundation model from scratch	Outside team SE mandatory scope
Unsupervised cloud control of hardware	Safety boundary of tabletop lab

2. System at a Glance

VISTA is a Robotics & AI Education Platform connecting schools, teachers and students with structured STEM learning and tabletop robot experimentation. School administrators manage classrooms and authorize robot resources; teachers prepare lessons, publish assignments and review student submissions. Students create versioned robot programs, submit work, access permitted experiment activities and receive feedback. The platform uses a React web application, a NestJS Education API, Supabase authentication and PostgreSQL, with separate AI integration and local robot runtime components. Its core value is traceable, role-scoped learning from assignment to program, experiment evidence and assessment. At Review 2, the Education API supports schematic experiment requests; physical robot execution remains a target requiring end-to-end verification. The architecture and integration boundaries are documented in Report 4 (SDD), C4 System Context and Container diagrams.

3. Key Design Decisions

Exactly 5 decisions. Each must have one owner.

ID	Decision	Options considered	Chosen option and rationale	Owner	Related FR / diagram
DD-01	Scoped API data access	Direct Supabase browser writes vs guarded NestJS	NestJS Education API + Prisma with membership guard	Lê Nguyễn Quang Đức	FR-01 / C4
DD-02	JWT tenant authorization	RLS-only vs app guards	Supabase JWT and organization/classroom guards	Lê Nguyễn Quang Đức	FR-01 / WF-01
DD-03	Robot entitlement allocation	Live reservation vs school allocation	Atomic classroom authorization separate from runtime	Lê Nguyễn Quang Đức	FR-02 / WF-01
DD-04	Experiment boundary	Coupled hardware calls vs separate runtime	Schematic path + separate local Python runtime	Vũ Anh Tuấn	FR-05 / WF-03
DD-05	Assessment and evidence	Embedded binary vs metadata reference	Store durable refs and teacher assessment separately	Phạm Viết Lợi	FR-06 / WF-04

4. Top Risks

ID	Risk	Impact	How we verify or mitigate	Owner
R-01	Physical runtime integration unverified	High	Separate schematic request from hardware execution and collect tested E2E evidence	Vũ Anh Tuấn
R-02	Cross-school authorization/data leak	High	Integration tests for JWT/membership/classroom scope	Lê Nguyễn Quang Đức
R-03	Diagram states and ERD constraints may differ from code	High	Validate each state against enum/service and FK/unique constraints	Trần Tiến Đạt

5. Known Gaps

Mismatches between the registration form, SRS, SDD and code. Include every High finding marked Kept or Planned in Form BM02.

ID	Gap	Location	Plan	Owner	Due
G-01	Physical robot E2E not verified	Education API / runtime	Keep as planned; test controlled robot workflow	Vũ Anh Tuấn	Before Defense 1.1
G-02	Proposed state machine labels	PDF state pages 8–11	Match exact enums and tests	Trần Tiến Đạt	Before BM02
G-03	Current approved registration unknown	BM03	Compare signed Review 1 version to v1.2	Lê Nguyễn Quang Đức	Before submission

6. Proposed Mandatory Features

Within the scope in section 1. Acceptance criteria must be testable with a clear pass/fail result.

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
FR-01	Authentication and role-scoped access	School Admin/Teacher/Student	Requests require valid user, membership, scoped authorization; cross-org access is denied	TC-FR-01	Partial / verify
FR-02	Classroom robot allocation	School Admin	Only entitled robot can be allocated to same-org classroom; replacement leaves one allocation	TC-FR-02	Partial / verify
FR-03	Course and lesson delivery	Teacher	Authorized teacher can assign published learning resources to assigned classroom	TC-FR-03	Partial / verify
FR-04	Program version and submission	Student	Own program version attached to own classroom assignment/attempt; other users cannot submit for learner	TC-FR-04	Partial / verify

7. Non-Functional Requirements

Maximum 6 rows. Each NFR needs a metric, a threshold and a measurement method.

ID	Quality attribute	Metric	Threshold	Measurement method
NFR-01	Authorization isolation	Cross-tenant unauthorized access	0 successful unauthorized cases	Run role/organization-scope negative integration tests
NFR-02	Core API correctness	Mandatory workflow main-path tests	100% pass by Defense 1.1	Run API integration tests with seeded fixtures
NFR-03	Demo reliability	Main/exception walkthrough	100% main; >=80% exception by Defense 1.1	Run staged demo cases with recorded outcomes

8. Ownership Map

One row per member. Each member owns 2–3 parts and at least one main workflow (Appendix A1).

Member	Role	Owned parts (2–3)	Main workflow owned	Key decisions / PRs
Lê Nguyễn Quang Đức	Leader	Auth/RBAC, school/robot, CI	WF-01	DD-01–03
Trần Tiến Đạt	Member	Learning/assignments, version/submission, UML	WF-02	Submission states
Vũ Anh Tuấn	Member	Experiment API, robot runtime, safety	WF-03	DD-04
Phạm Viết Lợi	Member	Assessment, feedback, AI metadata	WF-04	DD-05

Appendix A – Defense 1.1 Targets

A1. Main Workflows

Minimum number = number of team members. A valid workflow involves at least 2 actors or at least 3 mandatory features, and changes the state of at least one main entity.

ID	Workflow	Actors	Trigger	Business outcome	Features	Entity states	Owner
WF-01	School allocation	School Admin / Teacher	Classroom needs robot	Scoped resource authorized	FR-01/02	Allocation replaced; no runtime state	Lê Nguyễn Quang Đức
WF-02	Learning & assignment	Teacher / Student	New assignment	Versioned learner submission	FR-03/04	Published / submitted (verify labels)	Trần Tiến Đạt
WF-03	Experiment	Student / Runtime	Learner requests schematic	Pending request; physical target unverified	FR-05	pending in code; rest proposed	Vũ Anh Tuấn
WF-04	Teacher review	Teacher / Student	Submission available	Assessment/ feedback recorded	FR-06	Code-dependent review states	Phạm Viết Lợi

A2. Demo Scenarios

Per workflow: at least 1 main scenario and 1 exception scenario. Run on staging with seeded data.

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
S-01	WF-01	Main	Authorized school and entitled robot	Allocate eligible robot	Current class allocation persisted	Lê Nguyễn Quang Đức
S-02	WF-01	Exception	Different organization	Cross-org allocation attempt	Denied; no write	Lê Nguyễn Quang Đức
S-03	WF-02	Main	Teacher/student class membership	Publish and submit program version	Authorized submission persisted	Trần Tiến Đạt
S-04	WF-02	Exception	No class membership	Read/submit from foreign class	Denied	Trần Tiến Đạt
S-05	WF-03	Main	Valid request	Create schematic experiment	Pending request recorded	Vũ Anh Tuấn
S-06	WF-03	Exception	Unsupported run mode	Create physical via Education API	Rejected without physical actuation	Vũ Anh Tuấn

A3. Functional Test Case Target

Source	Count	Minimum per item	Subtotal
Mandatory FR	6	3 each	18
Main workflows	4	1 each	4
State transitions	Not yet verified	1 each	TBD from actual state enums
NFR	3	1 each	3

Source	Count	Minimum per item	Subtotal
TOTAL	TBD	Await verified transitions	>=25 plus state transitions

Automated in CI: at least 50% of the committed total.